EXPERIMENT TO UNDERSTAND LINEAR REGRESSION FOR A GIVEN DATA SET

PROGRAM:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Salary_data.csv')
print(df)

df.info()

df.dropna(inplace=True)

df.info()

print(df.describe())

features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=0)

from sklearn.linear_model import LinearRegression
model=LinearRegression()
```

```
model.fit(x_train,y_train)


print(model.score(x_train,y_train))
print(model.score(x_test,y_test))
print(model.coef_)
print(model.intercept_)


import pickle
pickle.dump(model,open('SalaryPred.model','wb'))


model=pickle.load(open('SalaryPred.model','rb'))


yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)


print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

OUTPUT:

```
  YearsExperience  Salary
0               1.1   39200
1               1.3   42100
2               1.5   44800
3               2.0   47200
4               2.2   50100
5               2.9   54800
6               3.0   56600
7               3.2   59100
8               3.7   62400
9               3.9   64600
10              4.0   67100
11              4.1   68900
12              4.5   72400
```

```
13              4.7   74800
14              5.0   78100
15              5.2   80600
16              5.3   83500
17              5.9   87200
18              6.0   89900
19              6.8   95100
20              7.1   97800
21              7.9  103500
22              8.2  107200
23              8.7  112400
24              9.0  116000
25              9.5  120500
26              9.6  123800
27             10.0  127500
28             10.3  131200
29             10.5  135000
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
       YearsExperience         Salary
count         30.00000      30.000000
mean           5.57000   83113.333333
std            2.92211   28988.734237
min            1.10000   39200.000000
25%            3.32500   59925.000000
50%            5.10000   79350.000000
75%            8.12500  106275.000000
max           10.50000  135000.000000
0.9965480200876051
0.9986651018711702
[[9917.57201802]]
[27760.44923723]
```

```
Enter Years of Experience:  5

Estimated Salary for 5.0 years of experience is [[77348.30932733]]:
```